

# Multi-Bottleneck Fairness for High-Precision Congestion Control

A Coexisting Switch-Minimal Mode for Multi-Bottleneck Max-Min Fairness without PFC

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## Abstract

HPCC (SIGCOMM '19) achieves near-optimal flow completion time for single-bottleneck RDMA workloads via INT-based precision congestion control with no per-flow switch state. We show its design has a structural fairness failure on *multi-bottleneck* paths — increasingly common in cross-pod and disaggregated datacenter traffic: a flow crossing several shared bottlenecks is starved by  $\sim 2\times$  relative to single-bottleneck competitors, and fails entirely once the per-hop INT record overflows its hop budget — a cap inherited from HPCC's original wire format. The collapse is winner-take-all, RTT-independent, and survives HPCC's own multi-rate mode and INT compression (PINT). Five representative sender-only corrections fail to close the gap, and we give the structural reason: at the target-utilization equilibrium every flow holds its rate, and the link's flow count is not observable in INT, leaving only additive redistribution that is orders of magnitude too slow (we claim empirical evidence, not formal impossibility). We argue the fair-share signal must live at the switch, and offer HPCC-FS not as a repair of HPCC but as an *alternative operating mode on the same INT fabric* for traffic that requires multi-bottleneck max-min fairness with zero PFC: one 64-bit header field and one scalar of state per egress port carry an RCP-style fair rate whose fixed point is the max-min share, reached *without ever counting flows*. On contention benchmarks the mode drives the multi-bottleneck penalty to  $\approx 1.0\times$  across bottleneck counts, asymmetric flow sizes, staggered starts, and ECMP hash seeds, with zero PFC; on allreduce-style coflows — where the job is gated by its slowest, multi-bottleneck flows — it cuts job completion time by  $\sim 20\%$  at every ECMP seed. We state explicitly when the mode should *not* be selected (mixed short-flow workloads) and where fabric-scale contention does not form. The mechanism is RCP; our primary contribution is the *diagnosis* and the demonstration that the fair-share signal belongs in the network, not the endpoint. Stock HPCC remains byte-for-byte unchanged alongside.

## 1 Introduction

HPCC [16] and its derivatives represent the state of the art in datacenter RDMA congestion control. Their design philosophy is unusually clean: the switch stamps raw per-hop state (transmitted bytes, queue length, link rate) into INT headers; the sender computes a precise utilization estimate and applies a multiplicative-increase, multiplicative-decrease update toward a target utilization

$\eta$ . The switch is stateless per flow; the wire format adds only a few bytes per hop; and on the canonical single-bottleneck benchmarks HPCC matches or beats DCQCN, TIMELY, and DCTCP on flow completion time and queueing.

Modern datacenter workloads, however, increasingly involve traffic that traverses *multiple contended links* in series: cross-pod ML allreduce, disaggregated memory and storage, and multi-tenant flows that share aggregation and core layers. Whether HPCC's design still delivers fair, low-latency service in these settings has not been carefully measured.

We address that question with a deterministic, oracle-grounded methodology. Our contributions:

- (1) **An empirical characterization** of HPCC's multi-bottleneck behaviour using a parameterized parking-lot benchmark and a fluid max-min oracle (§2). That a load-based scheme inherits some multi-bottleneck unfairness is folk knowledge; what the characterization adds is mechanism and magnitude that the folk claim does not predict: the failure is *winner-take-all* (the long flow collapses to  $\sim 0.4$  Gbps against  $\sim 23$  Gbps competitors, a  $\sim 1.9\times$  FCT penalty at  $N = 2-4$ ,  $3.5\times$  for small flows), *RTT-independent* (equalizing RTTs does not move it), robust to flow-size asymmetry and start staggering, immune to HPCC's own per-hop multi-rate mode and to INT compression (PINT), and compounded by a *wire-format* failure — beyond four bottlenecks the per-hop INT record overflows its 5-hop budget and the control input is corrupted outright.
- (2) **A negative result for sender-only fixes** (§3). We implement five sender-side variants of HPCC ranging from a  $k$ -aware additive boost to a global share-aware update rule, and show all five fail to close the gap. We give a structural explanation: at the multi-bottleneck equilibrium every flow holds its current rate (because the bottleneck utilization sits at  $\eta$ , making the multiplicative term  $\approx 1$ ), and a sender cannot compute the strong multiplicative correction needed without knowing the link's flow count  $N$ .
- (3) **HPCC-FS, a switch-minimal alternative operating mode for fairness-critical multi-bottleneck traffic** (§4). We do not position HPCC-FS as a repair of HPCC's control law; it is a *second service mode on the same INT fabric*, intended to be selected for the requirement profile HPCC does not serve — max-min fairness across multiple bottlenecks with

lossless, PFC-free operation — while stock HPCC continues to serve the classes it is best at. On contended flows each switch port runs a standard RCP [10] fair-rate update on aggregate load and queue, advertises that single rate in the INT header, and the sender adopts the path-minimum. The switch addition is minimal — one uint64 field in a new INT header mode and one scalar of state per egress port; the full deployment delta is slightly larger (a new sender ACK handler, and, in our prototype, rate-only operation with the per-flow window cap disabled). The mechanism is RCP, not a new control law: our contribution is showing *where* the fair-share signal must live, and that offering it as a coexisting mode is cheap — the existing HPCC code path (cc\_mode 3) is byte-for-byte unchanged.

- (4) **An evaluation showing HPCC-FS achieves the goal (§5):** max-min penalty  $\approx 1.005\times$  at  $N = 4$ , zero PFC events, robust across asymmetric workloads, generalizing to a tree-fabric and to a  $k = 4$  ECMP fat-tree — and, on allreduce-style coflows where the slowest flow gates the job, a  $\sim 20\%$  JCT reduction at every ECMP hash seed (§5.11).

Our framing positions the negative result (sender-only designs are insufficient in our experiments) as a *motivation* for the positive contribution (a coexisting, switch-minimal mode that meets the fairness-with-zero-PFC requirement on our benchmarks). To our knowledge, this is the first quantitative evidence that fair-share computation is hard to place purely at the sender in INT-style precision congestion control; we do not claim a formal impossibility (see Section 3).

*Artifacts.* The implementation, the topology/workload generators, the max-min oracle, and deterministic scripts that regenerate the figures (make\_figures.py) and the ablation/sensitivity/baseline tables (run\_sensitivity.py) from source are available in our artifact-review repository.<sup>1</sup>

## 2 Background and Motivation

### 2.1 HPCC, briefly

HPCC’s per-hop utilization estimate for hop  $i$  is

$$u_i = \frac{txRate_i}{C_i} + \frac{\min(qlen_i, qlen_i^{prev}) \cdot maxRate}{C_i \cdot W} \quad (1)$$

where the first term reflects bandwidth utilization and the second is a queue-drain term that keeps standing queues bounded. In single-rate mode the sender takes  $U = \max_i u_i$ , smooths it over the base RTT, and applies

$$new\_rate = \begin{cases} curRate / \max c + R_{AI} & \max c \geq 1 \\ curRate + R_{AI} & \text{otherwise} \end{cases}$$

with  $\max c = U/\eta$ . The window  $W$  is sized to  $m_{win} \cdot rate / NIC\_line$  under  $var\_win$ .

### 2.2 The multi-bottleneck parking lot

The *parking lot* is a classic congestion-control fairness benchmark [6, 13]. The name is an analogy: a long boulevard (the long flow’s path)

is crossed by a series of side streets, and at each intersection a different car (a single-link cross flow) merges on and exits. Because every link is equally contended, the *max-min fair* allocation [6] gives every flow an equal share, so the long flow should run exactly as fast as each cross flow despite traversing many contention points rather than one. The parking lot therefore isolates the multi-bottleneck fairness question in its simplest form: any deviation of the long flow’s rate from the cross flows’ is pure multi-bottleneck unfairness, with no confounding asymmetry in link capacity or hop count of the competitors.

We construct a parameterized parking-lot benchmark with  $N$  inter-switch bottleneck links (25 Gbps each) and 100 Gbps host links (Figure 1). One “long” flow traverses all  $N$  bottlenecks; one “cross” flow traverses each single bottleneck. Flows are 50 MB by default and start simultaneously at  $t = 0$ . With two flows sharing each 25 Gbps link, the max-min fair rate is 12.5 Gbps for every flow. All experiments are deterministic: each configuration produces byte-identical artifacts across repeated runs (verified).

*Metric.* We compare each flow’s actual flow completion time (FCT) against an idealized **fluid max-min oracle** — a continuous bandwidth-sharing model with no protocol dynamics (no slow start, no queueing, no convergence delay) in which the bandwidth is allocated max-min fairly at every instant. We compute the allocation by *progressive filling* [6]: all active flows raise their rate in lockstep until some link saturates; the flows crossing that link are frozen at their current rate; and the process repeats on the remaining flows until every flow is frozen. This is the textbook algorithm for max-min fair rates. Because flows complete and depart over time, we run it *event-driven* — recomputing the allocation at each flow completion — so the oracle FCT correctly accounts for the bandwidth freed as short flows finish. The oracle is therefore the best FCT any rate-allocation scheme could achieve under max-min fairness, and is exact for our fluid topologies. The headline metric is the **unfairness ratio**  $\max_{\text{path}} FCT / \text{mean}(\text{short-path FCT})$ ; the oracle’s value is 1.0 for equal-size, simultaneous flows.

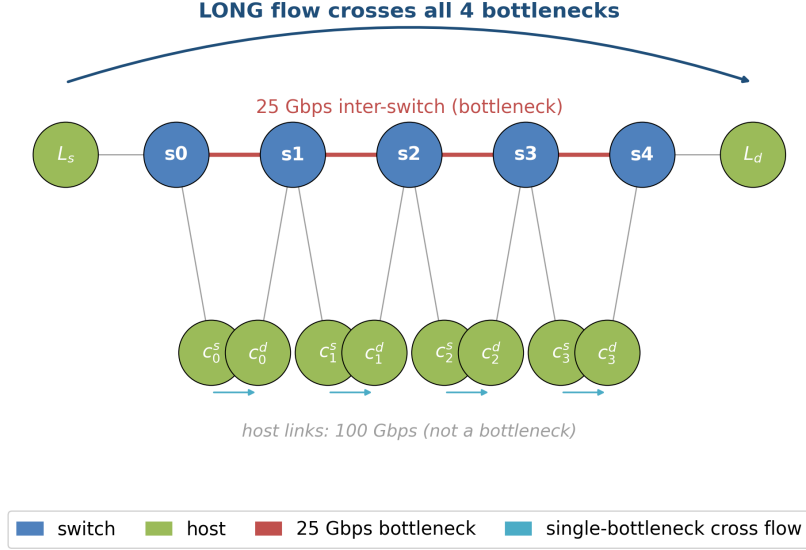
### 2.3 Findings F1–F4: characterizing the gap

*Magnitude.* Stock HPCC’s unfairness ratio at  $N \in \{2, 3, 4\}$  is 1.90, 1.92, 1.96: the multi-bottleneck flow’s FCT is roughly twice that of single-bottleneck competitors. **There is a non-monotonic regime change at  $N \geq 5$** , where the long flow becomes favored (0.51 $\times$ ); this turns out to be an INT-overflow artifact at  $\maxHop = 5$  and we exclude  $N \geq 5$  from the rest of this paper’s stock comparisons (Figure 2).<sup>2</sup>

*Robustness.* Perturbations across flow size and start time give penalties of 1.81 $\times$ –3.53 $\times$  (Figure 4): stable at 1.8–2.0 $\times$  and *worsening* dramatically (3.5 $\times$ ) for small multi-bottleneck flows — the latency-sensitive RPC / control case. Even an established long flow gets displaced when cross flows arrive.

<sup>1</sup><https://github.com/UNOCourseDemo/hpcc-fs>

<sup>2</sup>The 5-hop cap is HPCC’s original wire format, not an artifact of our re-port: the reference simulator hard-codes  $\maxHop = 5$  in its INT header [2], consistent with the paper’s 42-byte worst-case INT overhead (five 8-byte hop records). Conversely, a parking-lot path at  $N \leq 4$  traverses at most five switches — *within* the INT budget — so the retained  $N = 2$ –4 measurements are structurally free of the overflow.

Parking lot at  $N=4$ : one LONG flow + one CROSS flow per bottleneck

**Figure 1: Parking-lot benchmark at  $N = 4$ . The long flow  $L_s \rightarrow L_d$  crosses every bottleneck; each cross flow  $c_i^s \rightarrow c_i^d$  traverses exactly one. Host links are not bottlenecks.**

*HPCC's own multi-rate mode.* Multi-rate HPCC (per-hop EWMA, sender takes  $\min_i R c_i$ ) helps but does not fix the gap:  $1.44/1.50/1.75\times$  at  $N = 2/3/4$  — still growing with bottleneck count.

*RTT-equalization rules out RTT unfairness.* Sweeping the bottleneck link delay while holding  $N$  fixed, the penalty is essentially flat at  $\sim 1.9\times$  across cross-flow RTTs spanning  $10\times$  the long flow's RTT (including cases where cross flows have *longer* RTT than the long flow). The multi-bottleneck penalty is **not** RTT unfairness.

*Trace diagnosis.* A per-flow throughput trace at  $N = 4$  (Figure 5 left) shows the qualitative failure: the long flow collapses to  $\sim 0.38$  Gbps while the four cross flows hold  $\sim 23.2$  Gbps each for the entire 18 ms contention window; only after the cross flows complete does the long flow take the pipe. There are zero PFC events and zero loss. The collapse is “winner-take-all”: at every shared link the single-bottleneck cross flow wins locally and the long flow is squeezed.

### 3 Why Sender-Only Fixes Are Insufficient

A natural first design instinct — and the one our project explored exhaustively — is to keep all state and computation at the sender, since HPCC's value proposition is its stateless switch design. We tried five sender-only correction variants behind a new `mb_mode` config knob, on top of unchanged HPCC. Each is a one- or two-line change in `UpdateRateHp`; the `mb_mode = 0` path is byte-identical to stock HPCC (verified).

#### 3.1 The five variants

- (1)  **$k$ -aware additive increase** (`mb_mode 1`). The flow counts congested hops  $k$  from INT and scales  $R_{AI}$  by  $\gamma \cdot k$ . Result

**Table 1: Sender-side ablation: best penalty at  $N = 4$  for each variant.**

| mb_mode | mechanism                               | best penalty |
|---------|-----------------------------------------|--------------|
| 0       | stock HPCC                              | $1.96\times$ |
| 1       | $k$ -aware AI ( $\gamma = 50$ )         | $1.63\times$ |
| 2       | debiased blend ( $\lambda = 1$ )        | $1.95\times$ |
| 3       | responsibility-weighted MD              | $1.89\times$ |
| 4       | $k$ -th-root MD                         | $1.92\times$ |
| 5       | global share-aware AI ( $\gamma = 20$ ) | $1.71\times$ |

at  $N = 4$ : penalty drops from 1.96 to  $1.63\times$  at  $\gamma = 50$ , still nowhere near 1.0.

- (2) **Debiased max/mean blend** (`mb_mode 2`).  $U_{\text{eff}} = (1-\lambda) \max u_i + \lambda \text{mean}_{u_i \geq \eta} u_i$ . Result: penalty barely moves ( $1.95\times$  at  $\lambda = 1$ ), revealing that per-hop  $u_i$  are nearly equal in our setting — there is no significant max-bias to debias.
- (3) **Responsibility-weighted MD** (`mb_mode 3`). The flow's multiplicative decrease is scaled by its measured share  $\text{own\_rate}/\text{txRate}_{\text{btl}}$  at the bottleneck. Result:  $1.89\times$  — almost no effect, because the MD branch barely fires at the  $\eta$ -hold equilibrium.
- (4)  **$k$ -th-root MD** (`mb_mode 4`).  $\text{new\_rate} = \text{curRate} \cdot (1/\max c)^{1/k}$ . Result:  $1.92\times$ .
- (5) **Global share-aware additive increase** (`mb_mode 5`). Applied to *every* flow:  $R_{AI}^{\text{eff}} = 2(1 - \text{share})\gamma R_{AI}$  when congested. Result:  $1.71\times$  at  $\gamma = 20$ .

### 3.2 The structural reason

Two observations explain the pattern.

*The  $\eta$ -hold equilibrium.* Once the contended links settle at the target utilization  $\eta$ , HPCC's multiplicative term  $1/\max c \approx 1$ , so the rate update is dominated by holding. Every flow holds its current rate; if the dominant flows held a high rate at startup, they keep it. Additive nudges (modes 1, 5) take many tens of thousands of RTTs to redistribute meaningfully because  $R_{AI}$  is small compared to line rate. MD-side modifications (modes 3, 4) almost never fire because the multiplicative-decrease branch rarely activates at the hold point.

*The information asymmetry.* Strong redistribution requires moving the rate *multiplicatively* toward each flow's max-min fair share  $C/N$ . A sender knows  $C$  (from INT) and its own rate, but not  $N$  — the link's flow count. The only  $N$ -free sender mechanism is AIMD-style additive nudging, which we have just shown is too slow.

This is precisely why RCP [10] and XCP [13] put the fair-share computation in the switch. Our results are consistent with the same conclusion for the HPCC family: across the five representative sender-side designs we evaluated, none converges to the max-min allocation under the  $\eta$ -hold dynamics. We present this as empirical evidence, not a formal impossibility result — a tight lower bound on what sender-only INT can achieve is left to future work.

## 4 HPCC-FS Design

The empirical and structural arguments of §3 imply that the fix must (a) make the *dominant* flows yield, not just nudge the victims, and (b) place a fair-rate *signal* at the switch, since the sender cannot recover the fair share from a load signal alone. We do not have the switch count flows and divide; instead it runs a standard RCP control law on aggregate load and queue (below) whose fixed point is the max-min fair rate  $C/N$  — the switch converges to the fair share without ever materializing  $N$ . We take the most minimal mechanism consistent with these constraints, packaged as an *operating mode* selected per traffic class alongside stock HPCC on the same INT transport (§4.3).

*Background: RCP.* The Rate Control Protocol (RCP) [10] is a router-assisted congestion control in which each link advertises a *single* rate  $R$  to *all* flows crossing it, holding no per-flow state. The key point is that  $R$  is not an aggregate budget for the link; it is the rate *each individual flow* should send at, such that when all of them do, the link is exactly full and shared equally. A router raises  $R$  when its link has spare capacity and lowers it when a queue forms; routers stamp  $R$  into passing packets, each flow sends at the minimum  $R$  along its path, and that rate is echoed to the source. A single flow adopting  $R$  therefore does not seize the link — it takes its share, because  $R$  has already been driven down by the load the other flows place on the same port. Concretely, with  $N$  flows each sending  $R$  the offered load is  $NR$ , and the controller drives  $NR \rightarrow C$  (idle link  $\Rightarrow R$  rises; standing queue  $\Rightarrow R$  falls), so at equilibrium  $R \approx C/N$  — the max-min (processor-sharing) fair share, tracked *without ever counting  $N$* : as flows arrive or leave,  $R$  self-adjusts (a lone flow's  $R$  rises to  $C$ ; a third arrival drops everyone to  $C/3$ ). RCP thus supplies exactly the missing ingredient from §3: a fair-share signal computed where the flow count is implicitly

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### Algorithm 1 HPCC-FS switch — per egress port $j$

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```

1: State  $R_j$  (fair rate),  $t_j$  (last update),  $B_j$  (txByte snapshot)
2: Const  $C_j$  link capacity;  $\alpha=0.4$ ,  $\beta=0.226$ ;  $d$  control interval ( $\approx$ 
   RTT);  $R_{\min}$ 
3: procedure ONDEQUEUE( $pkt, j$ )  $\triangleright$  every packet leaving port  $j$ 
4:   if  $R_j$  uninitialised then
5:      $R_j \leftarrow C_j/2$ ;  $t_j \leftarrow \text{now}$ ;  $B_j \leftarrow \text{txBytes}_j$   $\triangleright$  conservative
       start
6:   end if
7:   if  $\text{now} - t_j \geq d$  then  $\triangleright$  one control interval elapsed
8:      $y \leftarrow (\text{txBytes}_j - B_j) \cdot 8 / (\text{now} - t_j)$   $\triangleright$  observed load
9:      $q \leftarrow \text{queueBytes}_j \cdot 8$   $\triangleright$  backlog (bits)
10:     $R_j \leftarrow R_j \cdot (1 + (\alpha(C_j - y) - \beta q/d)/C_j)$   $\triangleright$  RCP update
11:     $R_j \leftarrow \text{clamp}(R_j, R_{\min}, C_j)$ 
12:     $t_j \leftarrow \text{now}$ ;  $B_j \leftarrow \text{txBytes}_j$ 
13:  end if
14:  if  $pkt.\text{fairRate} = 0$  then  $\triangleright$  first hop on the path
15:     $pkt.\text{fairRate} \leftarrow R_j$ 
16:  else
17:     $pkt.\text{fairRate} \leftarrow \min(pkt.\text{fairRate}, R_j)$   $\triangleright$  path-min
18:  end if
19:   $\text{txBytes}_j \leftarrow \text{txBytes}_j + \text{sizeof}(pkt)$ 
20: end procedure

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### Algorithm 2 HPCC-FS sender (receiver echoes the header unchanged)

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```

1: on QP-create:  $rate \leftarrow R_{\min}$   $\triangleright$  conservative start
2: procedure ONACK( $a$ )
3:   if  $a.\text{fairRate} = 0$  then return  $\triangleright$  no signal yet
4:   end if
5:    $rate \leftarrow \text{clamp}(a.\text{fairRate}, R_{\min}, \text{NICrate})$   $\triangleright$  adopt path-min
       fair share
6: end procedure

```

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observable (the link), not at the sender. HPCC-FS runs this control law once per egress port and carries its  $R$  in the INT header; a flow's path-minimum  $R$  yields network-wide max-min fairness, since a flow bottlenecked elsewhere contributes less load here and the loop hands the slack to the rest.

### 4.1 The mechanism

The entire protocol is Algorithms 1 and 2. The switch maintains one scalar fair rate per egress port and stamps the path-minimum into a single header field; the sender reads that field from the returning ACK and adopts it. No per-flow switch state, no per-hop record, no receiver changes.

*At the switch (Alg. 1).* Each egress port keeps a single scalar fair rate  $R_j$ , refreshed once per control interval  $d$  ( $\approx$  one RTT) by the standard RCP rule  $R \leftarrow R(1 + (\alpha(C - y) - \beta q/d)/C)$ , where  $y$  is the load observed over the last interval and  $q$  the queue backlog. The term  $\alpha(C - y)$  chases spare capacity (positive while the link is under-utilized, pushing  $R$  up) and  $\beta q/d$  brakes when a standing queue forms (pulling  $R$  down); they balance at  $y \approx C$ ,  $q \approx 0$ , which

is the  $R \approx C/N$  fixed point above. We use the RCP defaults  $\alpha = 0.4$ ,  $\beta = 0.226$  (untuned; the result is insensitive to them, §5.6). The port’s entire state is three numbers —  $R_j$ , the last update time, and a byte-counter snapshot — with *no* per-flow data.

*On the wire.* The FS INT mode (`IntHeader::FS`) carries one 64-bit field, `fairRate`; the serialized cost is 8 bytes per packet *independent of hop count* (vs. 8 bytes *per hop* for stock HPCC). Each switch min-aggregates  $R_j$  into this field (first hop initialises it), so the receiver sees the path-minimum fair rate and echoes the header back unchanged — no receiver changes.

*At the sender (Alg. 2).* The new ACK handler `HandleAckFs` reads `fairRate` from the returning ACK and sets the sending rate to it, clamped to  $[R_{\min}, \text{NICrate}]$ . That single assignment is the whole sender control loop.

## 4.2 Startup smoothing

Two cold-start details matter for clean operation. **Switch:** initialize  $R = C/2$  (the common-case two-flow fair share), not  $C$  — otherwise the first stamped rate is line rate. **Sender:** initialize `qp->m_rate` to `m_minRate` for `cc_mode 11`, not the NIC line rate, so the first RTT before any ACK has returned does not blast at line rate. A flow obtains a path-complete fair-rate signal in a *single* RTT — unlike stock HPCC, whose per-link INT estimate needs multiple rounds to reflect a multi-bottleneck path — but full convergence to the max-min allocation takes longer, as the per-port RCP estimate settles over several control intervals (Section 5:  $\sim 4$  ms, i.e.  $\sim 10^2$  RTTs at our  $21 \mu\text{s}$  base RTT). HPCC-FS’s advantage is therefore *fast, path-length-independent* convergence, not literal one-RTT convergence.

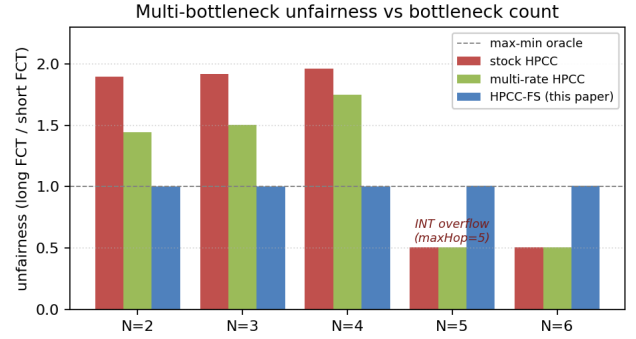
## 4.3 Additive integration and coexistence

HPCC-FS is `cc_mode == 11`. The mode-3 (HPCC) and mode-10 (PINT) wire formats and switch/sender code paths are completely untouched; we add a fourth case to (a) `IntHeader::Mode`, (b) its serialization, (c) the switch’s mode-dispatch, (d) the sender’s mode-dispatch, and (e) one branch in the validation driver. The driver also forces the QP window to 0 in FS mode (rate-only control), since `var_win` sizes the window from the NIC line rate rather than the bottleneck path BDP and would otherwise cap the multi-hop flow’s effective window well below what is needed at its fair rate. Because each mode has its own INT wire format and dispatch path, stock and FS traffic can in principle share a fabric with the mode chosen per traffic class — the intended deployment: latency-critical short-RPC classes stay on stock HPCC, collective and cross-pod classes elect HPCC-FS, and the FS class’s zero-PFC operation (§5.7) does not endanger the shared lossless fabric. Coexistence is established at the protocol and implementation level; our experiments run one mode per fabric, so mixed stock+FS classes *sharing links* remain future work.

## 5 Evaluation

### 5.1 Methodology

All experiments use the deterministic parking-lot benchmark of §2, driven by our `hpcc-validation.cc`. Flows are 50 MB unless noted; sim time is 200 ms; the optimized binary is used throughout. We compare against stock HPCC (`cc_mode 3, multi_rate`



**Figure 2: Multi-bottleneck unfairness vs bottleneck count  $N$ .** HPCC-FS is flat  $\approx 1.0$  across the entire sweep; stock and multi-rate HPCC fail at  $N \leq 4$  and break at  $N \geq 5$  (INT-overflow regime).

= false), stock HPCC multi-rate, HPCC-PINT, and the `mb_mode` ablation suite. The simulator is a refactored re-port of the original HPCC ns-3 stack onto a tagged ns-3.45 release; before using it here we verified that the re-port still reproduces the original paper’s baseline comparisons (HPCC vs. DCQCN, TIMELY, DCTCP, and PINT across WebSearch and FB-Hadoop workloads at 30–70% load on a 320-server fat-tree). The full reproduction record — per-run configurations, results, and analysis — ships with the artifact.

### 5.2 Headline: N-sweep

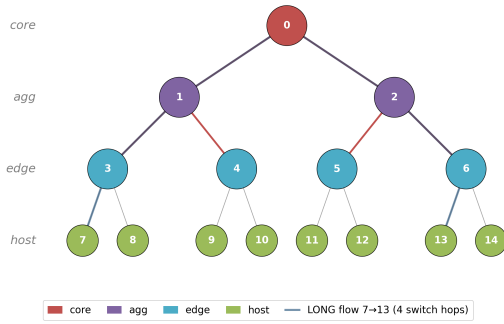
Stock and multi-rate HPCC are starved across  $N = 2$ –4 (penalty  $1.90$ – $1.96\times$  for stock,  $1.44$ – $1.75\times$  for multi-rate), then break entirely at  $N \geq 5$ , where the per-hop INT record overflows the `maxHop` cap (the apparent  $0.51\times$  there is the overflow artifact, not a real allocation). HPCC-FS is flat at  $1.003$ – $1.007\times$  across the entire sweep — hop-count-independent by construction, since its single fair-rate field does not grow with path length. HPCC-PINT, which carries one compressed utilization field, inherits HPCC’s load-based control and so still suffers the same  $\sim 1.8\times$  ( $1.77$ – $1.88\times$ ) multi-bottleneck penalty — confirming that the fix requires a fair-share signal, not merely a smaller INT footprint. All HPCC-FS runs have zero PFC events.

### 5.3 Structurally different topologies

*Tree fabric (3-tier, no ECMP).* A 3-tier tree with unique shortest paths (15 nodes; 25 Gbps inter-switch, 100 Gbps host; Figure 3a). Stock penalty  $1.36\times$ , HPCC-FS  $1.003\times$ , PFC = 0.

*$k = 4$  fat-tree (with ECMP).* The canonical fat-tree (20 switches + 16 hosts; 48 links; Figure 3b) with the simulator’s hash-based ECMP routing. Workload: 4 inter-pod flows (5-switch paths) plus 4 intra-pod different-edge flows (3-switch paths). We report two distinct quantities. (i) The *oracle-normalized penalty* — mean long-path slowdown divided by mean short-path slowdown, each measured against the max-min oracle — which removes the effect of ECMP load placement: stock  $1.34\times$ , HPCC-FS  $1.001\times$  (every contended flow lands at its oracle FCT). (ii) The *raw long/short FCT ratio*, which under HPCC-FS is still  $\sim 1.32\times$  — but this residual is

Tree fabric (3-tier, 15 nodes, unique shortest paths — no ECMP)



(a) Tree fabric — unique shortest paths, no ECMP.

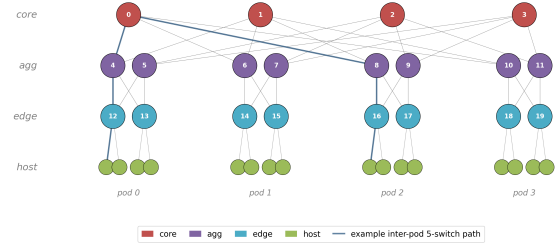
 $k = 4$  ECMP fat-tree: 4 cores, 8 aggs, 8 edges, 16 hosts(b)  $k = 4$  ECMP fat-tree (4 pods).

Figure 3: Two structurally different topologies used to test generalization beyond the parking lot. (a) 3-tier tree fabric, 15 nodes; the highlighted line is the long flow  $7 \rightarrow 13$ , crossing four switches. (b) Canonical  $k = 4$  fat-tree (4 cores, 8 aggs, 8 edges, 16 hosts in 4 pods); the highlighted line is a representative inter-pod 5-switch path.

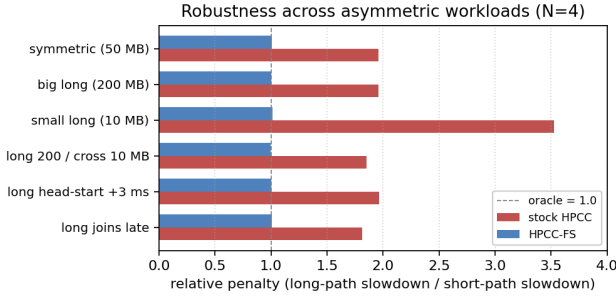


Figure 4: Robustness across asymmetric-size and staggered-start variants at  $N = 4$ . HPCC-FS is within  $1.0 \pm 0.012$  across all six scenarios, including the small-long worst case where stock reaches  $3.53\times$ .

not CC unfairness: ECMP hashes some flows onto under-loaded cores so they finish early, inflating the raw ratio independently of the congestion controller. PFC = 0 for both schemes. The point is that, once ECMP load placement is normalized away, HPCC-FS removes the multi-bottleneck penalty; the path-min fair-rate aggregation is path-invariant, so ECMP requires no design changes. *Hash-seed sensitivity.* Because a single deterministic run fixes one placement, we re-ran the workload across five ECMP hash seeds (an additive `ecmp_seed_offset` knob shifts every switch’s hash seed; the default preserves stock behavior). The stock raw ratio varies  $0.97\text{--}1.34\times$  across seeds with no change to the congestion controller — direct evidence that the raw spread is placement-dominated — while HPCC-FS’s raw ratio is at or below stock’s at every seed, and in the seed where placement produces real contention the ratio drops  $1.338\times \rightarrow 1.001\times$ . The oracle-normalized conclusion is therefore seed-robust (`run_ecmp_seeds.py` in the artifact).

Table 2: Window ablation: HPCC-FS rate-only (default) vs. with the per-flow window cap re-enabled.

| $N$ | FS rate-only (default) | FS window-on |
|-----|------------------------|--------------|
| 2   | $1.003\times$          | $1.50\times$ |
| 3   | $1.003\times$          | $2.00\times$ |
| 4   | $1.005\times$          | $2.49\times$ |

## 5.4 Robustness across asymmetric workloads

Across six perturbations (Figure 4), HPCC-FS holds within  $1.0 \pm 0.012$ . The worst stock case (small multi-bottleneck flow,  $3.53\times$  penalty) drops to  $1.012\times$ .

## 5.5 Ablation: rate-only is necessary

HPCC-FS runs rate-only in FS mode — the per-flow window cap is disabled (§4). Table 2 ablates this: re-enabling the window cap in FS mode degrades the penalty to  $1.5\text{--}2.5\times$ , *worsening with path length* (at  $N = 4$  it is worse than stock HPCC). The cause is that `var_win` sizes the window from the NIC line rate, not the smaller multi-hop bottleneck path, so the long flow is window-bound below its fair rate. Disabling the cap is thus a necessary part of the design, not an incidental change; the switch’s RCP rate already keeps queues bounded (zero PFC, §5.7).

## 5.6 Parameter sensitivity

A reasonable concern is that HPCC-FS’s RCP gains are tuned to the benchmark. They are not: we use the standard RCP defaults ( $\alpha = 0.4$ ,  $\beta = 0.226$ ) untouched. Sweeping each parameter *one at a time* at  $N = 4$  on the equal-size parking-lot (others at default), the penalty stays within  $[1.004, 1.008]\times$  with zero PFC throughout:  $\alpha \in [0.2, 0.6]$  ( $1.004\text{--}1.007\times$ ),  $\beta \in [0.1, 0.4]$  ( $1.005\times$ , flat), the startup fraction  $R_0/C \in [0.25, 1.0]$  ( $1.004\text{--}1.008\times$ ; even  $R_0 = C$  is PFC-free, because the sender’s `m_minRate` start absorbs the first-interval burst, §4), and `minRate`  $\in [100, 2000]$  Mb/s ( $1.005\times$ , flat). Within this benchmark family the RCP parameters are not load-bearing;

**Table 3: HPCC’s home turf (single-bottleneck incast). HPCC-FS holds far less queue than HPCC at every startup, and a moderate startup also matches or beats HPCC’s FCT. The same startup leaves the multi-bottleneck fairness headline unchanged ( $1.003\text{--}1.004\times$ , PFC = 0).**

|                          | mean FCT                       | tail FCT                       | peak queue   |
|--------------------------|--------------------------------|--------------------------------|--------------|
| HPCC                     | 258.6 $\mu$ s                  | 459.8 $\mu$ s                  | 143 KB       |
| HPCC-FS (startup 1 Gbps) | 297.5 $\mu$ s                  | 490.3 $\mu$ s                  | <b>64 KB</b> |
| HPCC-FS (startup 5 Gbps) | 271.2 $\mu$ s                  | 464.3 $\mu$ s                  | <b>49 KB</b> |
| HPCC-FS (startup 8 Gbps) | <b>242.0 <math>\mu</math>s</b> | <b>416.8 <math>\mu</math>s</b> | <b>38 KB</b> |

we do not claim the same insensitivity for arbitrary topologies, joint sweeps, or heterogeneous link speeds, which remain future work.

## 5.7 Queueing and PFC

With the smoothed startup (§4.2), HPCC-FS produces **zero PFC events** on every workload in the  $N$ -sweep and the robustness matrix. ECN marking remains; queue length is bounded (peak  $\approx 120$  KB at  $N = 4$ ); no flow loss.

## 5.8 Do-no-harm

The HPCC algorithm-validation smoke configuration (cc\_mode 3, single-bottleneck incast) passes all checks: 6/6 flows complete, 0 PFC, validation artifacts byte-identical to pre-modification. The stock cc\_mode 3  $N$ -sweep is bit-identical to the pre-change baseline. The HPCC-FS additions affect the cc\_mode 3 path only by adding new code; no existing function’s output for existing inputs is altered.

## 5.9 HPCC’s home turf: competitive FCT, lower queue

On contended flows HPCC-FS uses RCP-style switch-computed rates rather than HPCC’s per-hop control, so a fair question is what this costs on *single-bottleneck* workloads — HPCC’s home turf. We measure the validated incast benchmark (3 senders  $\rightarrow$  1 receiver over a 25 Gbps bottleneck; 12 flows, 100–400 KB), sweeping the sender’s startup rate (short flows begin at the startup rate and ramp until the first fair-rate ACK returns):

Two things stand out (Table 3). First, HPCC-FS holds *far less queue* than HPCC at *every* startup (38–64 KB vs 143 KB) — RCP’s  $\beta q/d$  term drains queue aggressively. Second, the only place HPCC leads is FCT at the conservative 1 Gbps startup ( $\sim 15\%$ ); raising the startup closes and then *reverses* that gap — at 8 Gbps HPCC-FS beats HPCC’s mean FCT (242 vs 259  $\mu$ s) at roughly one-quarter the queue, with zero PFC. The startup is a clean knob (queue rises again only past the point where the aggregate startup rate exceeds link capacity), and crucially it does *not* disturb the multi-bottleneck headline: at 8 Gbps the parking-lot penalty stays  $1.003\text{--}1.004\times$  ( $N=2, 3, 4$ ) with robustness variants  $1.002\text{--}1.009\times$ , all PFC = 0 (the same operating point as the headline runs). So on the home-turf workloads we measured, HPCC-FS’s fairness fix comes at *little-to-no FCT cost and a queue benefit* — not a trade-off.

We are careful not to overclaim: HPCC-FS still lacks HPCC’s per-hop, window-bounded reaction, so on rapidly-varying or adversarial-microburst workloads (which we did not evaluate) HPCC’s precision may matter; recovering that while keeping the fairness fix is what the hybrid overlay of §7 targets.

## 5.10 Per-flow rate trace, before vs after

The before-vs-after rate trace (Figure 5) is the cleanest qualitative result. Stock HPCC: the long flow collapses to  $\sim 0.4$  Gbps while the four cross flows hold  $\sim 23$  Gbps each for the full 18 ms contention; HPCC-FS: all five flows converge to  $\sim 12.5$  Gbps within  $\sim 4$  ms and stay locked at fair share.

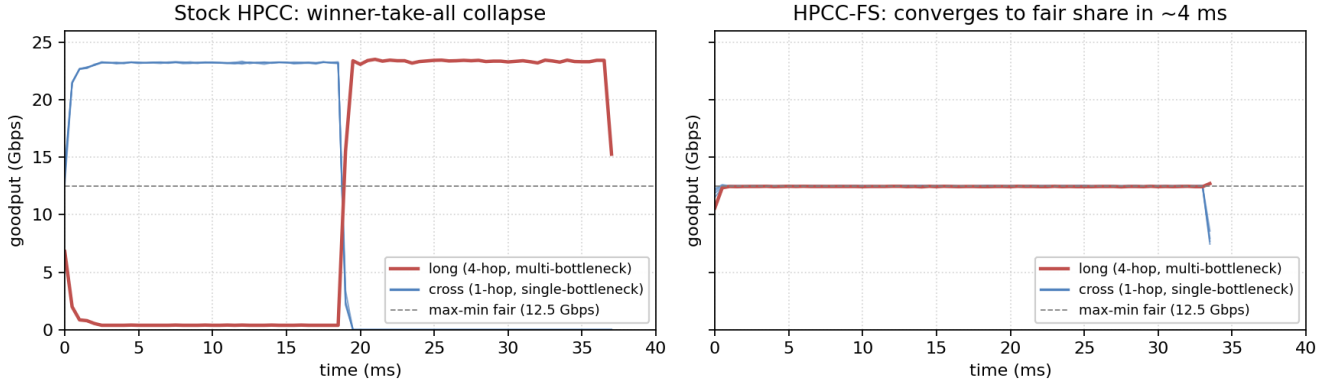
## 5.11 When max-min is the objective: allreduce-style coflow JCT

The workload class that motivates multi-bottleneck fairness is collective communication: a training step’s shard exchange or allreduce completes only when its *slowest* flow completes, so the job completion time is the coflow’s maximum FCT — and the slowest flows are precisely the cross-pod, multi-bottleneck ones that stock HPCC starves. We run a 16-flow ring allreduce chunk-exchange over all hosts of the  $k=4$  fat-tree (4 inter-pod, 4 intra-pod-different-edge, 8 same-edge segments; 50 MB each; JCT = max FCT), across the same five ECMP hash seeds. HPCC-FS cuts JCT by 19.5–22.7% (mean 21.4%) at *every* seed, and makes it nearly placement-invariant (17.9–18.0 ms across seeds, versus stock’s 22.3–23.3 ms) — equalized flows cannot be victimized by an unlucky hash. Adding eight 2 MB background flows leaves the JCT gain intact (+20.6% mean) and, in this mix, speeds the background flows up  $\sim 3\times$  (0.9–1.3 ms vs 2.2–4.3 ms): under winner-take-all dynamics whether a short flow wins or loses is a placement lottery (they lose here, they won in the WebSearch mix of §5.12), and max-min removes the lottery. The earlier shard-exchange workload of §5 shows the same job-level effect: JCT improves 9.8–26.5% across seeds (run\_allreduce\_jct.py).

## 5.12 Scope of the headline claim

The  $\approx 1.0\times$  headline is a statement about the *congestion-control component* of multi-bottleneck unfairness on contention benchmarks — not a claim of universal FCT improvement on production workloads. Three first-class limits: (i) *mixed short-flow workloads*: under a WebSearch size distribution, restoring max-min lowers multi-bottleneck flow slowdown by  $\sim 15\%$  but raises short-path mean slowdown by  $\sim 45\%$  (§7) — the headroom short flows previously enjoyed was subsidized by the long flow’s starvation, and workloads whose objective is short-flow FCT (the Homa/pFabric regime [4, 19]) may prefer that subsidy; max-min is the right target when multi-bottleneck flows matter — in particular when they gate a collective, where the subsidy inverts into a job-level tax (§5.11). In deployment this is a per-class *mode-selection* question, not a global replacement: classes that prefer the subsidy stay on stock HPCC; fairness-critical classes elect HPCC-FS (§4.3). (ii) *Fabric scale*: at  $k = 6, 8$  our scaling workload does not form persistent inter-pod contention (both schemes’ penalty  $< 1$ ), so the mode is exercised, but untested, where large-fabric contention would arise; a saturating workload is future work. (iii) *ECMP placement*: under ECMP an operator still observes a raw long/short spread ( $\sim 1.3\times$  at the





**Figure 5: Per-flow goodput at  $N = 4$ , stock HPCC (left) vs HPCC-FS (right). Stock collapses the long flow to  $\sim 0.4$  Gbps while cross flows hold  $\sim 23$  Gbps for 18 ms (winner-take-all). HPCC-FS converges all five flows to the max-min fair share (12.5 Gbps) within  $\sim 4$  ms.**

contended placement) that comes from hash load placement, not congestion control — HPCC-FS removes the CC component (§5), not the placement component, which is a routing/load-balancing problem.

## 6 Related Work

*Datacenter transport.* A long line of datacenter congestion control trades buffering for latency. DCTCP [3] reacts to ECN-marked standing queue; DCQCN [28] and TIMELY [18] bring rate-based control to lossless RDMA fabrics using ECN and delay respectively; and Swift [15] shows a carefully engineered delay signal scales to large fabrics. Receiver- and scheduling-driven designs such as Homa [19] and pFabric [4] instead prioritize short flows for low FCT. HPCC [16] marked a shift toward *in-network telemetry*: rather than infer congestion from a single scalar, the sender reads exact per-hop link load and sizes its window precisely, and On-Ramp [17] and PowerTCP [1] push the precise-signal philosophy further. Across this line the optimization target is overwhelmingly single-bottleneck FCT, tail latency, or queueing; *fairness across multiple shared bottlenecks*—our focus—receives comparatively little attention, and we show it is exactly where HPCC’s otherwise-precise control breaks.

*In-network telemetry.* INT [14] lets the data plane stamp per-hop state into packets on programmable pipelines, and HPCC was among the first to drive congestion control directly from it. The cost is header overhead, which later work minimizes: HPCC-PINT [26] compresses the per-hop record into a single probabilistic field, and Poseidon [25] restructures INT-based control for deployability on commodity hardware. HPCC-FS continues this minimal-telemetry direction but changes *what* the field carries: not a raw load sample to be interpreted by the sender, but an explicit fair *rate* computed at the switch and min-aggregated along the path—one 64-bit field, like PINT, yet a decision rather than a measurement.

*Explicit-rate and switch-assisted control.* Having the switch compute and return a sending rate is a classic idea. ATM ABR explicit-rate schemes such as ERICA [12] computed a fair share per port decades ago; XCP [13] decomposed switch feedback into separate

efficiency and fairness controllers; and RCP [10] reduced this to a single fair-rate scalar per link with no per-flow state, packet-stamped, with senders taking the path minimum. RCP is the closest mechanism to ours: HPCC-FS adopts its update rule but situates it as a *minimal fairness signal injected into an existing INT load-signaling system* rather than a standalone protocol, keeping HPCC’s single-rate sender and per-hop transport. A head-to-head against standalone RCP on the parking lot would compare *identical* control laws: on contended flows HPCC-FS’s data-plane dynamics *are* RCP’s by construction (the same per-port update, the same path-min aggregation, the same sender adoption), so the fairness numbers of §5 are, by that identity, also plain RCP’s numbers on this benchmark. What HPCC-FS adds over deploying RCP as a separate protocol is packaging, not control: one field on the INT transport a datacenter already runs, a byte-identical stock HPCC path, and per-cc\_mode coexistence — no second protocol stack, no new header on the wire. ABC [11] argues, for wireless links, that switches should compute the right control signal rather than expose raw load; we apply the same philosophy with minimal departure from HPCC’s datacenter design.

*In-switch fairness.* A parallel line enforces fairness inside the switch via scheduling or dropping. Fair queueing [9] and Deficit Round Robin [22] give each flow its own queue, while Core-Stateless Fair Queueing [24] and Approximate Fair Dropping [20] approximate fair bandwidth shares *without* per-flow queues—using a single estimated fair rate and probabilistic dropping, the same stateless-per-flow spirit as RCP and HPCC-FS. Programmable schedulers such as PIFO [23] and AIFO [27] make rich scheduling policies realizable at line rate. HPCC-FS differs in mechanism: rather than enforcing fairness by per-packet queueing or dropping, it *advertises* one fair rate per egress port and lets rate-controlled RDMA senders converge to it—one scalar of egress state, no per-flow tracking and no extra queues.

*Programmable-switch feasibility.* Reconfigurable match-action pipelines [8] and P4 [7] make the per-port computation HPCC-FS needs—an EWMA of egress load, a multiply-add rate update,



and stamping one header field—implementable on the same class of programmable switches that HPCC’s INT and Bolt [5] already assume.

*Closely related switch-assisted DC schemes.* Bolt [5] accelerates HPCC’s control loop with sub-RTT, per-flow switch assistance (Sub-RTT Control, Proactive Ramp-Up, Supply Matching), targeting tail latency and ramp-up. Its objective is orthogonal to ours: Bolt is not designed for, and does not claim, multi-bottleneck max-min fairness, and its feedback carries load/supply information rather than an explicit fair share — so the information asymmetry of §3 applies to its senders as well: a faster load-signal loop converges faster, but to the same equilibrium, since the fair share  $C/N$  is still not in the signal. A head-to-head (which would require re-implementing Bolt’s switch pipeline in our stack) would therefore measure reaction speed rather than the fairness mechanism this paper isolates; we leave it to future work. Annulus [21] uses a dual-control-loop design for traffic aggregates mixing WAN and datacenter flows—largely orthogonal to intra-DC multi-bottleneck fairness within a single class.

*Fairness foundations.* Max-min fairness and the progressive-filling characterization we use as our oracle are classical [6], and the parking-lot topology is a standard stress test for multi-bottleneck rate allocation. Our contribution is not a new fairness definition but the diagnosis that HPCC cannot realize this well-understood objective from sender-visible INT alone, and the demonstration that a single switch-computed fair rate restores it.

## 7 Discussion

*Relation to HPCC, and a hybrid path.* HPCC-FS is best understood not as a tweak to HPCC’s control law but as a demonstration of *where* the multi-bottleneck fix must live: a fair-share signal computed at the switch, which we instantiate with RCP. On contended flows it therefore replaces HPCC’s window-bounded, per-hop-INT control with an RCP-style rate, and inherits RCP’s character — fair and low-queue, but without HPCC’s aggressive single-flow precision (§5.9). The natural way to keep both is a *hybrid overlay*: run HPCC unchanged on every flow and use the switch’s fair rate only as a per-flow *ceiling*,  $rate = \min(rate_{HPCC}, fairRate)$ , so dominant flows are capped to their fair share (and yield bandwidth that HPCC’s own increase gives to the starved flow) while HPCC’s window and reaction are preserved. The open obstacle is the window-path mismatch behind our rate-only choice (§4): HPCC’s window is sized for the NIC, not the multi-hop bottleneck path, so a victim flow may stay window-bound below fair even after the ceiling frees bandwidth — the overlay likely needs a path-aware window. We leave this hybrid, which would retain HPCC’s precision *and* its fairness, to future work.

*INT hop limit and the non-monotonic flip.* The  $maxHop = 5$  limit in the INT-header layout is the source of the non-monotonic regime change in stock HPCC: paths longer than five switches overflow the INT record and the rate-control input is corrupted. HPCC-FS sidesteps this because its wire format carries one field independent of hop count, and our  $N = 5, 6$  measurements (§5) confirm it remains at penalty  $\approx 1.0\times$  in that regime.

*Mixed-size workloads.* In a preliminary test on the parking-lot under a Web-Search size distribution (30 flows, 483 B to 9.6 MB), HPCC-FS reduces multi-bottleneck flow mean slowdown by  $\sim 15\%$  but increases short-path mean slowdown by  $\sim 45\%$ . This is correct max-min behaviour — stock HPCC’s starvation of the long flow was inflating cross-link headroom that short flows benefited from — but it surfaces a workload-engineering question that max-min alone does not resolve. A defensible mixed-size evaluation needs per-class targets and is application-dependent; we leave it to follow-on work.

*Scaling to larger fat-trees.* The fat-tree generator runs at  $k = 6$  (45 switches, 36 hosts) and  $k = 8$  (80 switches, 64 hosts). At larger  $k$  the fabric has  $(k/2)^2$  cores, so inter-pod flows spread across uncongested core paths and persistent contention does not materialize for our scaling workload — both stock and HPCC-FS show penalty  $< 1$ , an ECMP load-distribution effect. HPCC-FS adds no overhead in this regime (0 PFC across all six runs). A workload designed to saturate larger fabrics with guaranteed inter-pod contention is open future work.

*Heterogeneous link speeds.* Our parking lot has uniform 25 Gbps bottlenecks. Mixed link speeds may interact with RCP’s  $\alpha(C - y)$  term in ways the symmetric case does not exercise. RCP itself is known to handle heterogeneity gracefully; we expect the same for HPCC-FS but have not verified.

*Multi-path / ECMP.* §5 shows HPCC-FS works under ECMP on a  $k = 4$  fat-tree with hash-based path selection — the path-min fair-rate aggregation is itself path-invariant. ECMP load imbalance does produce per-flow FCT spread, but this is a routing issue, not a CC issue.

*Hardware feasibility.* RCP’s update is a single per-port floating-point or fixed-point step with three multiplies and an add; it is well within the budget of programmable switches. The min-aggregation per packet is a 64-bit compare and conditional write.

*Limitations.* Three topology families (parking-lot, tree-fabric,  $k = 4-8$  fat-tree) and one in-sim baseline (HPCC-PINT); no head-to-head with hardware-assisted designs such as Bolt or PowerTCP — re-implementing them in this simulator is substantial, their objectives are orthogonal to multi-bottleneck max-min fairness, and a standalone-RCP head-to-head would compare identical control laws (§6); no testbed measurements; convergence is a few ms (not one RTT, §4); and the ECMP evaluation is a small controlled workload, though now swept across hash seeds (§5), not a realistic mixed-size + ECMP combination. The multi-bottleneck numbers come from a single (re-reported) simulator: the re-report reproduces the original paper’s baseline comparisons end-to-end (§5), and the retained  $N \leq 4$  regime sits within the original 5-hop INT budget by construction, but an independent-implementation cross-check of the parking-lot numbers remains future work. We did, however, verify that the result is insensitive to the RCP parameters (§5.6), so the absence of tuning is not a hidden confound.

## 8 Conclusion

We have shown that HPCC’s load-signaling INT design has a structural fairness failure in multi-bottleneck workloads — a real, robust, RTT-independent winner-take-all collapse with  $\sim 2\times$  FCT penalty

for the multi-bottleneck flow, up to  $3.5\times$  for small such flows. No sender-side correction we attempted (across five representative variants) closes the gap, because at the  $\eta$ -hold equilibrium additive nudges are too slow and the sender cannot compute its fair share without knowing the link's flow count. A minimal mechanism that suffices on our benchmarks is **one fair-rate field** in a new INT mode plus **one scalar of state per switch port** (and, in our prototype, disabling the per-flow window cap in FS mode — see Section 4). The resulting mode, HPCC-FS, is offered alongside stock HPCC rather than in its place — an option for traffic that requires multi-bottleneck max-min fairness with zero PFC on the same INT fabric. It achieves max-min fairness with penalty  $1.005\times$  and zero PFC at  $N = 4$  across all workloads we tested, converging in a few milliseconds independent of path length, while leaving the original HPCC code path unchanged.

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